
GNN-Based BERT for Understanding Context from Music

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Abstract

Music is a multi-layered signal containing complementary acoustic and semantic information. This project investigates a multimodal approach for music context understanding by representing audio tracks as graphs and combining graph-based acoustic representations with Transformer-based textual representations. Each track is divided into 5-second segments represented by 128-bin log-Mel features, with temporal and cosine-similarity edges forming the audio graph. A frozen DistilBERT encoder represents track titles, while GraphSAGE encodes the audio graph; their representations are integrated using both simple feature fusion and a cross-attention mechanism. Experiments on the FMA-small dataset using eight genre classes show that the Cross-Attention GNN+BERT model achieves 40.50% test accuracy, 41.18% Macro-F1, and 43.25% Macro AUC-PR, outperforming the individual DistilBERT, CNN, and GraphSAGE baselines. The results demonstrate that multimodal fusion provides the main performance improvement, while the additional benefit of cross-attention over simple fusion is comparatively modest.

1. Introduction

Music is a rich multimodal domain in which meaningful context is expressed through both acoustic and semantic information. Acoustic characteristics such as timbre, rhythm, harmony, and spectral patterns provide information about musical structure and genre, while textual information such as track titles, lyrics, tags, and captions provides complementary semantic context. A model that considers only one of these modalities may therefore miss important information required for reliable music understanding.

Recent advances in Graph Neural Networks (GNNs) provide a natural way to represent structured relationships within sequential and non-Euclidean data. Instead of treating an audio track as an independent collection of feature vectors, a graph can represent its temporal structure by connecting consecutive segments, while additional similarity edges can connect acoustically related segments. Message passing

over such a graph allows information from related musical segments to be aggregated before making a track-level prediction. In this project, GraphSAGE is used to learn representations from these audio segment graphs.

At the same time, Transformer-based language models have demonstrated strong capabilities for learning contextual representations from text. DistilBERT provides a computationally efficient alternative to full BERT while retaining much of its contextual representation capability (Sanh et al., 2019). In this work, DistilBERT is used to encode track titles and provide a semantic representation that complements the graph-based acoustic representation.

The central objective of this project is to investigate whether combining graph-based audio representations with Transformer-based textual representations can improve music genre classification compared with unimodal approaches. The study evaluates five models: a frozen DistilBERT title classifier, a convolutional neural network (CNN) audio baseline, a GraphSAGE audio model, a simple GNN-BERT fusion model, and a cross-attention GNN-BERT model. The experiments are conducted on the FMA-small dataset using eight genre classes.

The main contributions of this project are threefold. First, an audio graph representation is constructed from fixed-length log-Mel spectrogram segments using temporal and cosine-similarity relationships. Second, graph-based acoustic representations are combined with DistilBERT text representations through both simple feature fusion and cross-attention. Third, the models are compared using accuracy, Macro-F1, Micro-F1, and Macro AUC-PR, together with class-level analysis, t-SNE visualization, and individual case studies. The experimental results show that multimodal fusion provides the largest performance improvement, while cross-attention provides a smaller additional gain over simple feature fusion.

2. Related Work

2.1. Music Information Retrieval

Music information retrieval (MIR) focuses on extracting meaningful information from audio recordings, including genre, mood, tags, and other semantic properties. The Free Music Archive (FMA) provides a large collection of openly

available music together with metadata and standardized genre information, making it suitable for reproducible music classification experiments (Defferrard et al., 2017). In this project, the FMA-small subset is used to investigate genre classification from both acoustic and textual information.

2.2. Graph Neural Networks for Audio Representation

Graph Neural Networks provide a framework for learning representations from data with explicit relational structures. Unlike conventional neural networks that primarily operate on fixed grids or independent feature vectors, GNNs propagate information between connected nodes. GraphSAGE introduces an inductive neighborhood aggregation framework in which node representations are learned by aggregating information from local neighborhoods (Hamilton et al., 2017).

For music representation, graph structures can naturally capture relationships between different temporal segments of an audio track. Consecutive segments can be connected to represent temporal continuity, while similarity-based connections can capture relationships between acoustically related segments. This motivates the use of GraphSAGE to aggregate information across the segments of each track.

2.3. Transformer-Based Text Representation

Transformer architectures use self-attention to model relationships between elements in a sequence and have become widely used for contextual language representation (Vaswani et al., 2017). BERT introduced bidirectional Transformer-based pre-training for learning contextual representations that can be adapted to downstream language tasks (Devlin et al., 2019). DistilBERT is a smaller and computationally efficient BERT-family model obtained through knowledge distillation (Sanh et al., 2019).

In this project, DistilBERT is used as the BERT-based text encoder. Track titles are tokenized and passed through the frozen encoder to obtain contextual semantic representations. These representations provide complementary information to the acoustic features extracted from the audio graphs.

2.4. Multimodal Representation Fusion

Multimodal learning aims to combine information from different modalities so that complementary characteristics can contribute to a common prediction task. In the context of music understanding, acoustic information can describe properties such as timbre, rhythm, and spectral structure, while text can provide semantic information associated with the track.

A straightforward approach is to concatenate representations

from different modalities before classification. More expressive fusion mechanisms can instead allow one modality to condition its representation on information from another modality. Cross-attention provides such a mechanism by allowing a query representation from one modality to attend to feature representations from another modality. In this work, both simple feature fusion and cross-attention are investigated to determine whether explicit interaction between textual and acoustic representations improves genre classification.

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2.6. Summary of Related Work

The existing literature provides complementary foundations for the proposed approach. The FMA dataset provides a suitable benchmark for reproducible music classification experiments (Defferrard et al., 2017), while GraphSAGE provides an inductive framework for learning representations through neighborhood aggregation (Hamilton et al., 2017). Transformer-based models such as BERT and its compressed variant DistilBERT provide contextual representations for textual information (Devlin et al., 2019; Sanh et al., 2019). Building on these ideas, this project combines graph-based audio representations with DistilBERT-based textual representations and investigates whether cross-modal attention can improve music genre classification.

3. Dataset and Preprocessing

3.1. Dataset

This project uses the Free Music Archive (FMA) dataset, specifically the FMA-small subset, which contains 8,000

audio tracks together with genre metadata (Defferrard et al., 2017). The FMA-small subset is selected because it provides a manageable collection of publicly available music suitable for reproducible supervised learning experiments.

The classification task uses eight genre categories: Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, and Rock. The original FMA metadata provides the track identifiers, genre labels, and track titles used in the experiments. The official dataset split is preserved throughout the experiments to avoid introducing additional train–test contamination.

The final dataset contains 6,400 training tracks, 800 validation tracks, and 800 test tracks. The official split is balanced across the eight selected genres, with 800 training tracks, 100 validation tracks, and 100 test tracks per genre before graph-generation filtering.

3.2. Task Definition

The original project specification considers multimodal music understanding using both audio and textual information. In the implemented experiment, the primary supervised task is formulated as an eight-class music genre classification problem.

For each track, the input consists of an audio representation and a textual representation derived from the track title. The target variable is the corresponding genre label. Formally, for a track i , the input can be represented as

$$T_i = (X_i^{audio}, X_i^{text}, y_i), \quad (1)$$

where X_i^{audio} denotes the acoustic representation, X_i^{text} denotes the title representation, and $y_i \in \{0, \dots, 7\}$ denotes the genre label.

Although the broader project specification allows multi-label tagging and other music-context prediction tasks, the implemented experiment uses single-label eight-class genre classification because the available metadata consistently provides a primary genre label and track title for the selected subset.

3.3. Audio Preprocessing

Each audio track is resampled to 22,050 Hz before feature extraction. The audio signal is then divided into fixed-length segments of 5 seconds. Only complete 5-second segments are retained so that every graph node corresponds to the same temporal duration.

For each segment, a 128-bin log-Mel spectrogram is computed. The resulting time-frequency representation is summarized along the temporal dimension to obtain a fixed-dimensional feature vector of size 128. Thus, if a track

contains N complete five-second segments, its acoustic representation is expressed as

$$X^{audio} = [x_1, x_2, \dots, x_N], \quad x_i \in \mathbb{R}^{128}. \quad (2)$$

This representation provides a compact description of the spectral characteristics of each local audio segment while preserving the segment-level structure required for graph construction.

3.4. Audio Graph Construction

Each track is represented as a graph $G = (V, E)$, where each node corresponds to one complete five-second audio segment. The node feature matrix is therefore

$$X_G \in \mathbb{R}^{N \times 128}, \quad (3)$$

where N is the number of valid audio segments in the track.

Two types of edges are used to capture complementary relationships between segments. First, bidirectional temporal edges are added between consecutive segments to represent the sequential structure of the audio. Second, cosine-similarity edges are added between acoustically related segments. For each node, the two most similar other segments are selected according to cosine similarity.

The combination of temporal and similarity-based edges allows the graph to represent both local sequential continuity and non-local acoustic relationships. Consequently, the graph structure is not restricted to a simple chain and can connect segments that exhibit similar spectral characteristics even when they occur at different positions within the track.

Duplicate edges are removed during graph construction to maintain a deterministic graph representation. The resulting graph is stored using PyTorch Geometric’s graph data format, with node features, edge indices, genre labels, and track identifiers preserved for subsequent training and evaluation.

3.5. Text Preprocessing

The textual modality is represented using the track title associated with each FMA track. Titles are tokenized using the DistilBERT tokenizer with the pretrained model `distilbert-base-uncased` (Sanh et al., 2019).

A maximum sequence length of 32 tokens is used because the track titles in this dataset are substantially shorter than typical document-level inputs. Titles longer than the specified limit are truncated, while shorter titles are padded according to the tokenizer requirements.

The DistilBERT encoder is kept frozen during training. Its contextual representation is extracted from the encoded title and subsequently passed to a trainable classification or

fusion component. Freezing the encoder reduces computational requirements and ensures that the experiment primarily evaluates the contribution of the learned multimodal fusion architecture.

3.6. Feature Normalization

To avoid information leakage between the dataset splits, feature normalization is fitted using only the training graphs. A `StandardScaler` is incrementally fitted over the node features of the training graphs and then applied without refitting to the validation and test graphs.

For an individual feature dimension, the normalized value is computed as

$$\hat{x} = \frac{x - \mu_{train}}{\sigma_{train}}, \quad (4)$$

where μ_{train} and σ_{train} are estimated exclusively from the training data.

This procedure ensures that information from the validation and test sets is not used during preprocessing. The fitted scaler is stored and reused consistently across the GraphSAGE and multimodal experiments.

3.7. Graph Generation and Data Availability

The official train, validation, and test split assignments are retained throughout preprocessing. During graph generation, six training tracks were excluded because their audio files were missing, unreadable, or did not contain a complete five-second segment under the fixed-segment preprocessing procedure. The excluded track identifiers are 98565, 98567, 98569, 99134, 108925, and 133297.

After this filtering step, the final graph dataset contains 6,394 training graphs, 800 validation graphs, and 800 test graphs. The validation and test sets remain aligned with the official split, while the reduction in the training set is explicitly accounted for during multimodal representation alignment.

For reproducibility, representative graph files are included in the project repository, while the complete processed graph collection and generated textual embeddings are treated as derived project artifacts.

4. Methodology

4.1. Overall Architecture

The proposed system combines graph-based acoustic representation learning with Transformer-based textual representation learning for music genre classification. Each music track is represented through two complementary modalities:

an audio graph constructed from fixed-length audio segments and a textual representation obtained from the track title.

The overall architecture consists of three main components. First, GraphSAGE is used to encode the audio graph and produce a track-level acoustic representation. Second, a frozen DistilBERT encoder produces a contextual representation of the track title. Finally, the two modality-specific representations are combined using either simple feature fusion or a cross-attention mechanism before classification.

For a track i , the two modality-specific representations can be written as

$$h_i^{audio} = \text{GNN}(G_i, X_i), \quad (5)$$

and

$$h_i^{text} = \text{DistilBERT}(X_i^{text}), \quad (6)$$

where G_i and X_i denote the audio graph and its node features, respectively. The resulting representations are then combined to obtain a multimodal representation used for genre prediction.

4.2. GraphSAGE Audio Encoder

The acoustic representation is learned using GraphSAGE (Hamilton et al., 2017). GraphSAGE performs neighborhood aggregation so that each node can incorporate information from its connected neighbors. This is particularly suitable for the constructed music graphs because the edges represent both temporal continuity and acoustic similarity between audio segments.

Let $h_v^{(l)}$ denote the representation of node v at layer l . A GraphSAGE layer aggregates the representations of neighboring nodes and combines the aggregated information with the node’s current representation. The general update can be expressed as

$$h_v^{(l+1)} = \sigma \left(W^{(l)} \left[h_v^{(l)} \parallel \text{AGG} \left(\{h_u^{(l)} : u \in \mathcal{N}(v)\} \right) \right] \right), \quad (7)$$

where $\mathcal{N}(v)$ represents the neighborhood of node v , AGG is a neighborhood aggregation operation, $\parallel$ denotes concatenation, $W^{(l)}$ is a learnable transformation, and σ is a nonlinear activation function.

The implemented GraphSAGE encoder contains two GraphSAGE convolution layers. The first layer transforms the 128-dimensional input node features into a 128-dimensional hidden representation. ReLU activation and dropout are

applied after the first layer. A second GraphSAGE layer further propagates information across the graph.

After message passing, node representations are aggregated using global mean pooling to obtain a fixed-dimensional representation for the complete track:

$$h_i^{audio} = \text{MeanPool} \left(\{h_v^{(2)} : v \in V_i\} \right). \quad (8)$$

The resulting 128-dimensional vector represents the acoustic characteristics of the complete track.

4.3. DistilBERT Text Encoder

The textual modality is encoded using the pretrained `distilbert-base-uncased` model (Sanh et al., 2019). For each track, the title is tokenized and passed through the Transformer encoder. The resulting contextual representation is used as the semantic representation of the track.

The DistilBERT encoder is frozen during training. Therefore, its parameters are not updated using the genre-classification objective. Instead, the extracted 768-dimensional representation is passed to a trainable projection or classification component.

Let $h_i^{BERT} \in \mathbb{R}^{768}$ denote the representation obtained from DistilBERT. For multimodal fusion, this representation is projected into the same 128-dimensional feature space used by the audio encoder:

$$h_i^{text} = W_t h_i^{BERT} + b_t, \quad (9)$$

where

$$W_t \in \mathbb{R}^{128 \times 768}. \quad (10)$$

This projection makes the textual representation compatible with the graph-based acoustic representation.

4.4. Simple GNN-BERT Fusion

As a baseline multimodal fusion strategy, the GraphSAGE and DistilBERT representations are concatenated directly. The fused representation is defined as

$$z_i = h_i^{audio} \parallel h_i^{text}. \quad (11)$$

Since both representations have dimensionality 128 after projection, the resulting fused representation has dimensionality 256.

The fused representation is passed through a fully connected classification network consisting of a 256-to-128

linear layer, ReLU activation, dropout, and a final linear layer producing eight genre logits. The predicted class is obtained from the maximum output logit.

This simple fusion model provides an important baseline for evaluating whether combining acoustic and textual information is beneficial even without explicitly modeling interactions between the two modalities.

4.5. Cross-Attention GNN-BERT Fusion

The main multimodal architecture extends simple feature concatenation with cross-attention. The motivation is to allow the textual representation to selectively attend to different audio segments rather than treating all graph-level acoustic information equally.

After the second GraphSAGE layer, the node representations are retained as a sequence of segment-level features. Because different tracks contain different numbers of five-second segments, the variable-length node sequences are converted into a padded dense representation with an associated node mask.

The projected textual representation is used as the query of a multi-head attention mechanism, while the graph node representations are used as both keys and values. Formally, the projected textual representation is

$$q_i = W_t h_i^{BERT}, \quad (12)$$

and the audio node representations form the key and value sequences

$$K_i = V_i = [h_{i,1}, h_{i,2}, \dots, h_{i,N_i}]. \quad (13)$$

The cross-attention operation can then be expressed conceptually as

$$\text{Attn}(q_i, K_i, V_i) = \text{softmax} \left(\frac{q_i K_i^T}{\sqrt{d}} \right) V_i, \quad (14)$$

where d denotes the attention feature dimension.

The implementation uses four attention heads. The resulting attended textual representation is combined with the original projected text representation through a residual connection and layer normalization:

$$\tilde{h}_i^{text} = \text{LayerNorm} \left(\text{CrossAttn}(q_i, K_i, V_i) + h_i^{text} \right). \quad (15)$$

In parallel, a global mean pooling operation is applied to the GraphSAGE node representations to obtain a global acoustic

representation. The final multimodal representation is then constructed as

$$z_i = \tilde{h}_i^{text} \parallel h_i^{audio}. \quad (16)$$

The resulting 256-dimensional representation is passed through a fully connected classification head to predict the eight genre classes.

4.6. Classification Objective

All classification models are trained using the standard multiclass cross-entropy loss. Given the predicted logits $\hat{y}_i$ and the ground-truth class y_i , the loss is defined as

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \log \left(\frac{\exp(\hat{y}_{i,y_i})}{\sum_{c=1}^C \exp(\hat{y}_{i,c})} \right), \quad (17)$$

where N is the number of training examples and $C = 8$ is the number of genre classes.

The same primary classification objective is used for the CNN, GraphSAGE, simple GNN-BERT, and cross-attention GNN-BERT models. The DistilBERT baseline also uses the same genre classification objective, but only its trainable classification head is optimized while the pre-trained DistilBERT encoder remains frozen.

4.7. Model Variants

To evaluate the contribution of each component, five model configurations are compared:

1. **DistilBERT Title:** A frozen DistilBERT encoder followed by a trainable classification head using only track titles.
2. **CNN:** A convolutional neural network operating directly on the log-Mel spectrogram representation of the first complete five-second audio segment.
3. **GraphSAGE:** A graph-based audio model operating on all complete five-second segments and their temporal and similarity-based graph connections.
4. **Simple GNN-BERT:** Concatenation-based fusion of the GraphSAGE audio representation and the projected DistilBERT text representation.
5. **Cross-Attention GNN-BERT:** The proposed multimodal model in which the textual representation attends to segment-level GraphSAGE representations before classification.

This progression provides a controlled comparison from unimodal text and audio baselines to graph-based audio

modeling and finally multimodal fusion. It also allows the effect of simple fusion to be separated from the additional contribution of cross-attention.

5. Experimental Setup

5.1. Data Splits

All experiments use the official FMA-small train, validation, and test partitions. The original split contains 6,400 training tracks, 800 validation tracks, and 800 test tracks. The same partitions are maintained across the text, CNN, GraphSAGE, and multimodal experiments to ensure that model comparisons are performed on consistent evaluation sets.

During graph construction, six training tracks were excluded because their audio files were missing, unreadable, or did not contain a complete five-second segment under the fixed-segment preprocessing procedure. Therefore, the final graph-based training set contains 6,394 graphs, while the validation and test sets contain 800 graphs each.

The exclusion is restricted to the training set and does not modify the official validation or test partitions. Track identifiers are retained throughout preprocessing so that graph representations and textual embeddings can be aligned correctly for multimodal training.

5.2. Training Configuration

All neural network models are trained using the AdamW optimizer. Unless otherwise specified, a learning rate of 10^{-3} and weight decay of 10^{-4} are used. Dropout with a probability of 0.3 is applied in the trainable neural network components to reduce overfitting.

The GraphSAGE model is trained for 30 epochs using a cross-entropy classification objective. Model selection is performed using validation accuracy, and the checkpoint corresponding to the best validation performance is retained for final test evaluation.

The CNN baseline is trained for 8 epochs using the same optimizer and learning-rate configuration. The CNN operates on the first complete five-second audio segment available for each track.

The DistilBERT text baseline is trained for 20 epochs. The pretrained DistilBERT encoder is frozen, and only the trainable classification head is optimized. This reduces the number of trainable parameters and makes the experiment computationally feasible on the available CPU-based environment.

Both multimodal models are trained for 20 epochs. The simple GNN-BERT fusion model and the cross-attention GNN-BERT model use the same basic optimizer configu-

ration so that differences in performance can be attributed primarily to the fusion strategy.

5.3. Model Selection

Validation performance is used to select the best checkpoint for each model. The model state corresponding to the highest validation accuracy is saved and subsequently evaluated once on the held-out test set.

This procedure prevents the test set from influencing model selection. The test set is used only after training and checkpoint selection have been completed.

For the cross-attention GNN-BERT model, the best validation performance is obtained at epoch 8. The corresponding checkpoint is used for the final evaluation. The best simple GNN-BERT checkpoint is obtained at epoch 16, while the best GraphSAGE checkpoint is obtained at epoch 11.

5.4. Evaluation Metrics

Because the task is an eight-class single-label classification problem, several complementary metrics are reported.

Accuracy. Accuracy measures the proportion of test examples for which the predicted genre is identical to the ground-truth genre:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i = y_i), \quad (18)$$

where $\mathbb{I}(\cdot)$ is an indicator function.

Macro-F1. Macro-F1 calculates the F1 score independently for each class and then averages the resulting scores:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c. \quad (19)$$

This metric gives equal importance to every genre and is therefore useful for identifying whether a model performs well only on a small number of classes.

Micro-F1. Micro-F1 aggregates the contributions of all classes before calculating the F1 score. For the balanced single-label classification setting used in this project, Micro-F1 is numerically equivalent to accuracy.

Macro AUC-PR. Macro AUC-PR measures the area under the precision-recall curve separately for each class and then averages the class-level scores. It provides an additional evaluation of the quality of the predicted probability

distributions, particularly when class-level prediction confidence differs across genres.

5.5. Baseline Comparisons

Five models are evaluated to understand the contribution of the different modalities and architectural components:

1. **DistilBERT Title Baseline:** uses only the track title as textual input.
2. **CNN Audio Baseline:** uses a convolutional network operating on the first complete five-second log-Mel segment.
3. **GraphSAGE Audio Model:** uses all available complete five-second segments represented as nodes in the constructed audio graph.
4. **Simple GNN-BERT Fusion:** combines the GraphSAGE and DistilBERT representations through direct concatenation.
5. **Cross-Attention GNN-BERT:** combines the two modalities using multi-head cross-attention between the textual representation and segment-level graph representations.

The baseline progression is designed to answer three questions. First, how informative is the textual modality by itself? Second, does graph-based audio modeling provide an advantage over a conventional CNN baseline? Third, does combining acoustic and textual representations improve performance beyond either modality individually, and does cross-attention provide an additional benefit over simple concatenation?

5.6. Implementation Details

The experiments are implemented using PyTorch and PyTorch Geometric. Graph construction and graph-level batching are handled using PyTorch Geometric, while the pre-trained DistilBERT tokenizer and encoder are provided by the Transformers library.

The audio processing pipeline uses a sampling rate of 22,050 Hz and 128-bin log-Mel features. Graph nodes correspond to complete five-second segments, and graph edges combine bidirectional temporal connections with the top-two cosine-similarity relationships for each node.

The GraphSAGE hidden dimension is 128. The multimodal models project the 768-dimensional DistilBERT representation into the same 128-dimensional space. The cross-attention module uses four attention heads. The final multimodal representation has 256 dimensions after concatenating the attended textual representation with the global graph representation.

Table 1. Test-set performance of the evaluated models. Best results are shown in bold.

Model	Acc.	Macro-F1	Micro-F1	AUC-PR
DistilBERT	26.62	24.99	26.62	25.51
CNN	31.50	29.40	31.50	32.48
GraphSAGE	37.13	35.88	37.13	38.72
Simple GNN-BERT	40.38	40.78	40.38	42.71
Cross-Attn. GNN-BERT	40.50	41.18	40.50	43.25

All experiments are conducted using a CPU-based training environment. To improve reproducibility, the preprocessing scaler, model checkpoints, representative graph files, generated embeddings, and evaluation outputs are saved as project artifacts.

6. Results

6.1. Overall Model Comparison

Table 1 summarizes the performance of all five evaluated models on the held-out test set. The results show a clear progression from unimodal baselines toward multimodal architectures.

The frozen DistilBERT title model achieves 26.62% accuracy and 24.99% Macro-F1, indicating that track titles alone provide useful but limited information for predicting musical genre. The CNN baseline performs better, reaching 31.50% accuracy and 29.40% Macro-F1. This indicates that acoustic information contained in the log-Mel representation provides stronger genre-related information than the title-only textual representation in this experimental setting.

The GraphSAGE model further improves performance to 37.13% accuracy and 35.88% Macro-F1. Its improvement over the CNN baseline suggests that representing the audio as a graph can capture relationships between different temporal segments that are not explicitly modeled by the conventional convolutional baseline. However, this comparison should be interpreted with care because the two models use different amounts of audio context: the CNN baseline uses only the first complete five-second segment, whereas GraphSAGE processes all complete five-second segments available for each track.

The multimodal models achieve the strongest overall performance. Simple GNN-BERT fusion increases accuracy to 40.38% and Macro-F1 to 40.78%. The cross-attention GNN-BERT model achieves the best overall result, with 40.50% accuracy, 41.18% Macro-F1, 40.50% Micro-F1, and 43.25% Macro AUC-PR.

These results support the main hypothesis that combining acoustic and textual representations is more effective than

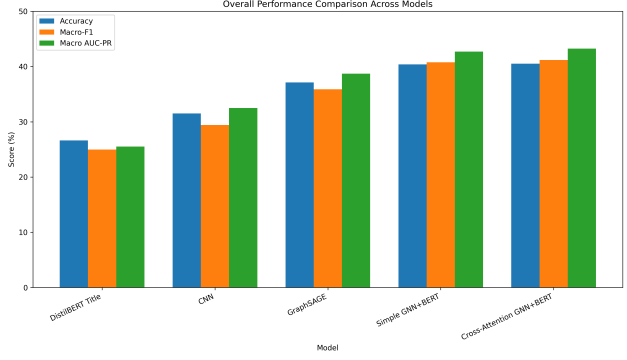


Figure 1. Overall test-set performance comparison of the evaluated DistilBERT, CNN, GraphSAGE, Simple GNN-BERT, and Cross-Attention GNN-BERT models.

Table 2. Ablation of multimodal fusion strategies.

Fusion Method	Accuracy	Macro-F1	Micro-F1	AUC-PR
Simple Fusion	40.38	40.78	40.38	42.71
Cross-Attention	40.50	41.18	40.50	43.25

using either modality independently. The improvement from GraphSAGE to simple GNN-BERT is particularly notable, indicating that textual information provides complementary information beyond the graph-based acoustic representation

Figure 1 provides a visual comparison of the overall performance of the five evaluated models.

6.2. Fusion Ablation

To examine whether explicit cross-modal interaction provides an advantage over direct concatenation, the two multimodal fusion strategies are compared in Table 2.

Cross-attention provides a modest improvement over simple concatenation. Accuracy increases by 0.12 percentage points, while Macro-F1 increases by approximately 0.41 percentage points. Macro AUC-PR increases by approximately 0.54 percentage points.

The relatively small difference between the two fusion mechanisms suggests that the primary benefit comes from combining the two modalities rather than from the specific attention mechanism itself. Nevertheless, cross-attention achieves the best value across all reported metrics, indicating that allowing the textual representation to selectively interact with segment-level acoustic features provides a useful, although limited, additional benefit.

6.3. Class-Level Performance

The class-level results reveal substantial variation in genre recognition performance. The Cross-Attention GNN-BERT model achieves its strongest F1 scores on Hip-Hop (64.71%), International (57.75%), and Rock (53.76%). These classes appear to contain acoustic and/or semantic characteristics that are comparatively easier for the multimodal representation to distinguish.

In contrast, Folk (21.77%), Pop (21.05%), and Experimental (26.34%) remain more challenging. These classes may contain substantial intra-class variation or acoustic similarities with other genres. In particular, the relatively low performance on Folk and Pop indicates that genre boundaries are not always clearly represented by the available audio and title information.

The GraphSAGE model also performs particularly well on Hip-Hop and Rock, with F1 scores of 60.87% and 55.77%, respectively. This suggests that acoustic structure alone can provide strong evidence for some genres. However, multimodal fusion improves several classes where the unimodal models are less reliable.

6.4. Confusion Analysis

The confusion matrices provide additional insight into the classification behavior of the models. The GraphSAGE model correctly identifies 70 of the 100 Hip-Hop test examples and 58 of the 100 Rock examples. However, substantial confusion occurs among genres such as Folk, Experimental, Instrumental, and International.

The cross-attention GNN-BERT model correctly identifies 60 Electronic, 27 Experimental, 27 Folk, 55 Hip-Hop, 33 Instrumental, 54 International, 18 Pop, and 50 Rock test examples. The model therefore maintains relatively strong recognition for Hip-Hop, International, and Rock while improving the balance of predictions across several other classes.

The confusion patterns also demonstrate that genre classification is not uniformly difficult across the eight classes. For example, Folk is frequently confused with Experimental and Instrumental, while Pop receives predictions from several other categories. Such patterns suggest that the learned representation captures broad acoustic and semantic similarities rather than perfectly separable genre boundaries.

6.5. Main Finding

Overall, the experimental results provide evidence that multimodal representation learning is beneficial for the implemented music genre classification task. The best-performing model combines GraphSAGE-based audio representations with DistilBERT-based textual representations

through cross-attention.

Compared with the title-only DistilBERT baseline, the cross-attention model improves test accuracy from 26.62% to 40.50%, corresponding to an absolute improvement of 13.88 percentage points. Macro-F1 increases from 24.99% to 41.18%, an improvement of 16.19 percentage points.

Compared with the GraphSAGE audio-only model, multimodal cross-attention improves accuracy by 3.37 percentage points and Macro-F1 by 5.30 percentage points. These improvements indicate that the title representation provides complementary information that is not fully captured by the acoustic graph.

At the same time, the relatively small difference between simple fusion and cross-attention indicates that the major performance gain is attributable to multimodal combination itself. The attention mechanism contributes an additional but comparatively modest improvement.

7. Representation Analysis

7.1. t-SNE Visualization

To investigate how the learned representations organize the different music genres, t-SNE visualizations are generated for the audio, textual, and fused representations of the 800 test tracks (van der Maaten & Hinton, 2008). t-SNE is applied independently to each representation using a two-dimensional embedding with perplexity 30, PCA initialization, automatic learning rate selection, and random seed 42.

Figure 2 compares the resulting two-dimensional representations. The audio representation corresponds to the 128-dimensional GraphSAGE track-level embedding, while the textual representation corresponds to the projected 128-dimensional DistilBERT embedding. The fused representation corresponds to the 256-dimensional representation obtained by combining the attended textual representation with the global acoustic representation of the cross-attention GNN-BERT model.

The audio representation exhibits stronger organization for several acoustically distinctive genres, particularly Hip-Hop and Rock. This observation is consistent with the relatively high class-level F1 scores achieved by GraphSAGE for these genres. However, substantial overlap remains among genres such as Folk, Experimental, Instrumental, and International, indicating that their acoustic characteristics are not completely separable in the learned embedding space.

The textual representation shows a different organization from the audio representation. Since the text modality is based only on track titles, its semantic information is relatively limited compared with lyric- or caption-based rep-

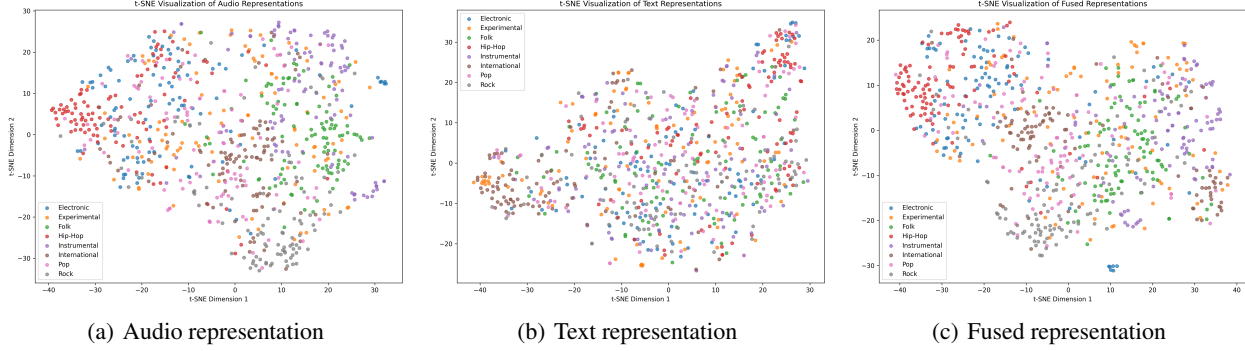


Figure 2. t-SNE visualization of the test-set representations. The audio representation is obtained from the GraphSAGE encoder, the text representation is obtained from the projected DistilBERT embedding, and the fused representation combines acoustic and textual information through cross-attention.

representations. Nevertheless, some genre-related structure is present, suggesting that titles can provide useful semantic cues even without additional textual context.

The fused representation provides a more informative joint representation by combining the complementary information from both modalities. Several genre groups become more structured after fusion, although considerable overlap remains. This behavior is consistent with the quantitative results: the multimodal models outperform both the text-only and audio-only baselines, but the classification problem remains challenging for several genre categories.

The visualization therefore supports the quantitative finding that the two modalities capture complementary information. At the same time, the remaining overlap demonstrates that multimodal fusion does not produce perfectly separable genre representations and that additional information, such as lyrics, richer musical structure, or larger-scale audio representations, could potentially improve performance.

8. Case Studies

To complement the aggregate evaluation metrics, three test-set examples are examined in detail to understand how the individual modalities and multimodal fusion behave on individual tracks. The selected examples represent different prediction patterns: a case where multimodal fusion improves over both unimodal models, a case where the acoustic graph provides particularly strong evidence, and a case where textual information corrects an acoustic-only prediction.

8.1. Case Study 1: Helicopter de Cristal

The first example is track 1066, titled “Helicopter de Cristal,” whose ground-truth genre is Experimental. The DistilBERT model predicts International as its most probable class with a probability of 39.48%, while GraphSAGE predicts Instru-

mental with a probability of 26.48%.

The simple GNN-BERT model instead predicts Experimental with a probability of 29.41%, while the cross-attention GNN-BERT model also predicts Experimental, with a probability of 25.88%. Although the confidence of the multimodal model is relatively modest, both multimodal variants identify the correct genre whereas neither unimodal model produces the correct top prediction.

This example illustrates the complementary nature of the two modalities. The title alone does not provide sufficient information for the correct prediction, and the acoustic graph also favors a different genre. Combining the two sources of information allows the multimodal models to recover the correct Experimental label.

8.2. Case Study 2: New Wave Hippies

The second example is track 4108, titled “New Wave Hippies,” with Rock as the ground-truth genre. The DistilBERT model predicts Experimental with a probability of 44.43%, showing that the title representation alone does not reliably identify the correct genre.

In contrast, GraphSAGE predicts Rock with a probability of 82.24%. The simple GNN-BERT model increases the Rock probability further to 94.34%, while the cross-attention model predicts Rock with a probability of 64.55%.

This case demonstrates a situation in which the acoustic modality provides strong evidence for the correct genre. The multimodal models preserve this useful acoustic information and outperform the text-only baseline. The simple fusion model achieves particularly high confidence for the correct Rock class, suggesting that the textual representation does not necessarily need to dominate the prediction when the acoustic graph already provides a strong genre signal.

8.3. Case Study 3: Mano De Dios

The third example is track 704, titled “Mano De Dios,” with International as the ground-truth genre. The DistilBERT model predicts International with a probability of 44.51%, whereas GraphSAGE predicts Rock with a probability of 61.49%.

The simple GNN-BERT model predicts International with a probability of 53.69%, and the cross-attention model also predicts International, with a probability of 50.79%.

This example demonstrates the complementary role of the textual modality. The acoustic graph alone favors Rock, but the title representation provides evidence supporting the correct International class. Both multimodal models successfully shift the prediction toward the correct genre.

8.4. Case Study Summary

The three examples demonstrate different ways in which multimodal fusion can contribute to genre prediction. In the first case, both unimodal models favor incorrect genres, while multimodal fusion identifies the correct class. In the second case, the acoustic graph provides the dominant signal and the multimodal models preserve the strong acoustic prediction. In the third case, the textual representation provides useful information that helps correct an acoustic-only error.

These examples support the interpretation that the acoustic and textual modalities provide complementary information rather than uniformly contributing in the same way for every track. The multimodal architecture can therefore benefit from either modality depending on the characteristics of an individual example. However, the case studies also show that multimodal fusion does not guarantee correct predictions, as the overall test performance remains substantially below perfect classification.

9. Discussion

9.1. Effect of Multimodal Fusion

The experimental results indicate that combining acoustic and textual information provides a consistent advantage over unimodal models. The DistilBERT title baseline achieves 26.62% accuracy, while the GraphSAGE audio model achieves 37.13%. When the two representations are combined through simple feature fusion, accuracy increases to 40.38% and Macro-F1 increases to 40.78%.

This improvement suggests that the two modalities capture complementary information. The acoustic graph represents spectral and temporal characteristics of the music, whereas the title representation provides semantic information that is not directly available from the audio features. Consequently,

the multimodal model can use information from one modality when the other modality is ambiguous or insufficient.

The improvement is also visible in the case studies. For example, “Helicopter de Cristal” is incorrectly classified by both unimodal models but correctly classified by both multimodal models. Similarly, the title representation helps correct the acoustic-only prediction for “Mano De Dios.” These examples provide qualitative evidence that multimodal fusion can compensate for weaknesses in individual modalities.

9.2. Contribution of Cross-Attention

Although cross-attention produces the best overall performance, its improvement over simple feature concatenation is relatively small. Accuracy increases from 40.38% to 40.50%, while Macro-F1 increases from 40.78% to 41.18%. Macro AUC-PR increases from 42.71% to 43.25%.

This result suggests that the main performance improvement comes from combining the modalities rather than from the additional complexity of the cross-attention mechanism itself. Simple concatenation already provides access to complementary acoustic and textual information, while cross-attention provides a more explicit mechanism for conditioning the textual representation on individual audio segments.

The relatively modest improvement may also be related to the limited textual information available in the experiment. The textual modality consists only of track titles rather than full lyrics, detailed tags, or natural-language captions. As a result, the semantic representation may not contain enough information for cross-attention to consistently identify highly informative audio segments.

Therefore, the results support cross-attention as a useful fusion mechanism, but they do not establish that cross-attention is substantially superior to simple fusion under the current experimental conditions.

9.3. Importance of Graph-Based Audio Representation

The GraphSAGE model outperforms the CNN baseline on all reported metrics. GraphSAGE achieves 37.13% accuracy, compared with 31.50% for the CNN baseline, while Macro-F1 increases from 29.40% to 35.88%.

One possible explanation is that the graph representation explicitly models relationships between different temporal segments of a track. Temporal edges preserve local continuity, while cosine-similarity edges allow acoustically related segments to exchange information even when they occur at different positions in the track. The resulting message-passing process can therefore integrate information from multiple parts of the audio.

However, the CNN and GraphSAGE comparison must be

interpreted carefully. The CNN baseline uses only the first complete five-second segment of each track, whereas GraphSAGE processes all complete five-second segments available for the track. Therefore, the observed improvement cannot be attributed exclusively to the graph structure. The GraphSAGE model also has access to a larger portion of the audio signal.

A more controlled future comparison would provide the CNN with the same complete set of audio segments or an equivalent track-level aggregation before classification. Such an experiment would isolate the contribution of graph-based relational modeling more directly.

9.4. Class-Level Behavior

The class-level results show that genre classification is substantially easier for some categories than for others. The cross-attention model performs particularly well on Hip-Hop, International, and Rock, with F1 scores of 64.71%, 57.75%, and 53.76%, respectively.

These results suggest that these genres contain comparatively distinctive combinations of acoustic and semantic characteristics in the selected dataset. In contrast, Folk, Pop, and Experimental remain difficult classes, with F1 scores below 30%.

The confusion patterns indicate substantial overlap between several of the challenging genres. Folk is frequently confused with Experimental and Instrumental, while Pop receives predictions from multiple other categories. Such errors are plausible because genre boundaries are not always acoustically discrete, and individual tracks can contain characteristics associated with multiple musical styles.

The class-level analysis therefore demonstrates why Macro-F1 is useful alongside accuracy. Accuracy alone can hide substantial differences in performance across individual genres, whereas Macro-F1 gives equal importance to each class.

9.5. Limitations

Several limitations should be considered when interpreting the results.

First, the textual modality is limited to track titles. The original project motivation considers richer textual information such as lyrics, tags, and captions, but the implemented experiment does not use these sources. Consequently, the textual representation may contain substantially less contextual information than a richer music-language representation.

Second, the DistilBERT encoder is frozen. Freezing the encoder reduces computational requirements and provides a controlled baseline, but it prevents the language representation from adapting to the music-domain classification task. Fine-tuning the Transformer could potentially produce more

task-specific semantic features.

Third, the CNN and GraphSAGE audio models receive different amounts of audio context. The CNN uses the first complete five-second segment, whereas GraphSAGE uses all complete five-second segments available for each track. Therefore, the GraphSAGE improvement should not be interpreted as a pure measurement of graph-structure effectiveness.

Fourth, six training tracks are excluded during graph construction because their audio files are missing, unreadable, or too short to provide a complete five-second segment. Although the official validation and test partitions remain unchanged, this reduces the effective training set for the graph-based and multimodal experiments.

Finally, the current experiment focuses on eight-class genre classification. It does not evaluate the broader multi-label tagging, emotion regression, or cross-modal retrieval objectives described in the original project motivation. These tasks would provide additional evidence about whether the proposed representation is useful for general music-context understanding.

9.6. Implications and Future Work

The results suggest several directions for future development. A richer textual modality could be introduced by incorporating lyrics, user-generated tags, or MusicCaps-style natural-language descriptions. Such information could provide substantially more semantic context than short track titles.

The audio representation could also be improved by using beat-synchronous segmentation, chroma features, learned spectrogram encoders, or longer temporal contexts. In addition, a controlled CNN comparison using the same complete set of audio segments as GraphSAGE would provide a more rigorous evaluation of the benefit of graph-based modeling.

Future work could also investigate fine-tuning the DistilBERT encoder and using stronger cross-modal architectures. Multi-layer cross-attention, bidirectional audio-text attention, or gated fusion could provide more flexible interaction between the modalities.

Finally, the framework could be extended beyond genre classification to multi-label music tagging, emotion prediction, and audio-text retrieval. These tasks would more directly evaluate the broader objective of understanding musical context rather than predicting a single genre label.

10. Conclusion

This project investigated a multimodal approach to music genre classification by combining graph-based acoustic rep-

representations with Transformer-based textual representations. Audio tracks were represented as graphs of five-second segments using 128-bin log-Mel features, temporal connections, and cosine-similarity relationships. GraphSAGE was then used to learn track-level acoustic representations, while a frozen DistilBERT encoder provided contextual representations from track titles.

The experimental results demonstrate that multimodal fusion provides a clear advantage over the individual modalities. The DistilBERT title baseline achieved 26.62% test accuracy and 24.99% Macro-F1, while the GraphSAGE audio model achieved 37.13% accuracy and 35.88% Macro-F1. Combining the two modalities through simple feature fusion increased performance to 40.38% accuracy and 40.78% Macro-F1. The proposed cross-attention GNN-BERT model achieved the best overall performance, reaching 40.50% accuracy, 41.18% Macro-F1, and 43.25% Macro AUC-PR.

The results also show that the main benefit comes from integrating complementary acoustic and textual information. Cross-attention provides a smaller additional improvement over simple concatenation, suggesting that explicit cross-modal interaction is useful but is not the primary source of the performance gain in the current experimental setting. The t-SNE analysis and individual case studies further support the observation that the two modalities capture different aspects of musical context.

Several limitations remain, particularly the use of short track titles instead of richer textual information, the frozen DistilBERT encoder, and the unequal audio context available to the CNN and GraphSAGE baselines. Future work could address these limitations by incorporating lyrics, tags, or natural-language music descriptions, fine-tuning the language encoder, using richer audio representations, and evaluating additional tasks such as multi-label music tagging, emotion prediction, and audio-text retrieval.

Overall, the study demonstrates the potential of combining graph-based audio modeling with Transformer-based textual representations for music understanding. The findings provide evidence that multimodal representations can improve genre classification by leveraging complementary information from both the acoustic structure and semantic context of music.

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